

AXERAI

— Products receive digital life

Connected Packaging Operating System

Technical & Product Document — WebAR experience, AI conversation, Inworld voice synthesis, and persistent ledger memory for physical products.

App-less WebAR

3-Layer Verify

Gemini Flash-Lite

Inworld TTS

Supabase Ledger

EXAMPLE PRODUCT CODE

AXE-327T5

TECHNOLOGY STACK

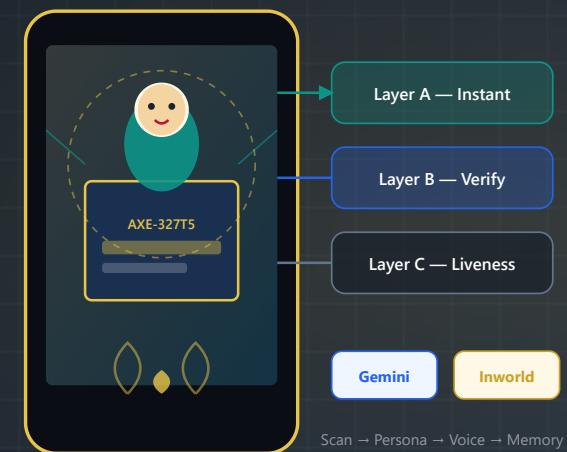
React · Vite · Three.js · MindAR

DOCUMENT VERSION

July 2026 Pilot

CLASSIFICATION

Confidential



 Contents

01	Executive Overview	07	AI Brain (Google Gemini)
02	Technology Stack	08	Voice Pipeline (Inworld API)
03	End-to-End User Journey	09	Speech-to-Text & Live Mic
04	Three-Layer Verification Architecture	10	Ledger, Memory & Token Metering
05	System Architecture Diagram	11	Deployment & Security
06	D3 Character — Animation & Lip Sync	12	Project Structure

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APP INSTALL REQUIRED

3

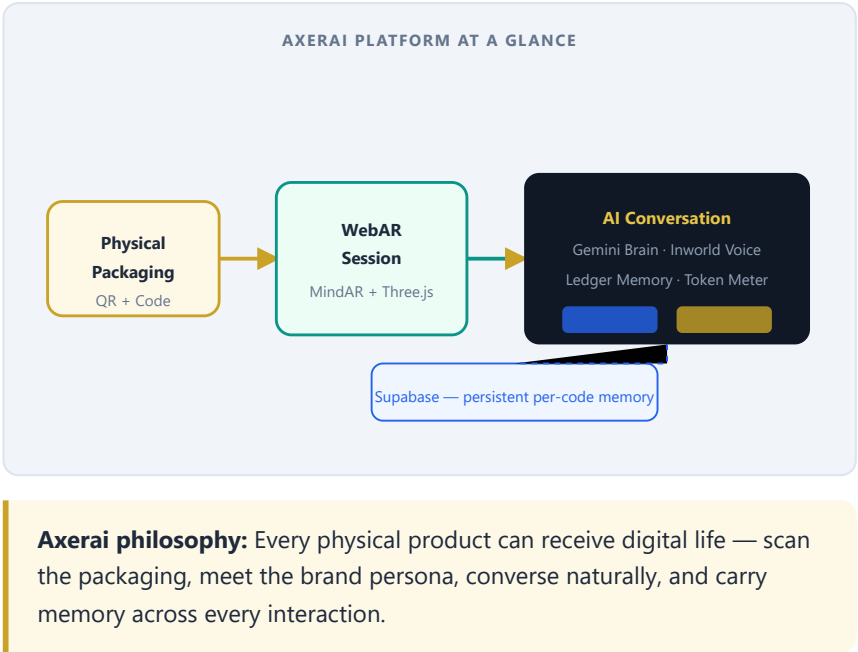
VERIFICATION LAYERS

2

ROLES PER CODE

FBX

3D PERSONA MODEL



1 Executive Overview

Axeraï transforms physical product packaging into an intelligent, conversational digital touchpoint — without requiring a native mobile application.

WHAT IT IS

AI-Native Packaging OS (MVP)

Consumer scans QR / image target on box → 3D persona anchors on product → natural voice conversation via Gemini + Inworld. Memory persists per product code across sessions.

WHAT IT IS NOT

Not a static QR website

Not a chatbot widget. Not a brochure. The packaging becomes a living brand ambassador with memory, verification, and voice — patent-pending 3-layer architecture.

2 Technology Stack

LAYER	TECHNOLOGY	ROLE
Frontend	React 19 + Vite 8 + Tailwind 4	UI, state, mobile shell
3D / AR	Three.js + MindAR	Image target tracking, persona anchor
Character	FBX (Mixamo rig + morph targets)	Idle/talk animations, mouth open/close
AI Brain	Google Gemini 3.1 Flash-Lite	Chat, verify vision, session summary
Voice TTS	Inworld API (cloned persona)	Brand voice output
STT	Browser Web Speech API (en-IN)	Live microphone input
Database	Supabase PostgreSQL	Ledger, memory, token logs
Hosting	Netlify + Serverless Functions	Static deploy + secure API proxy

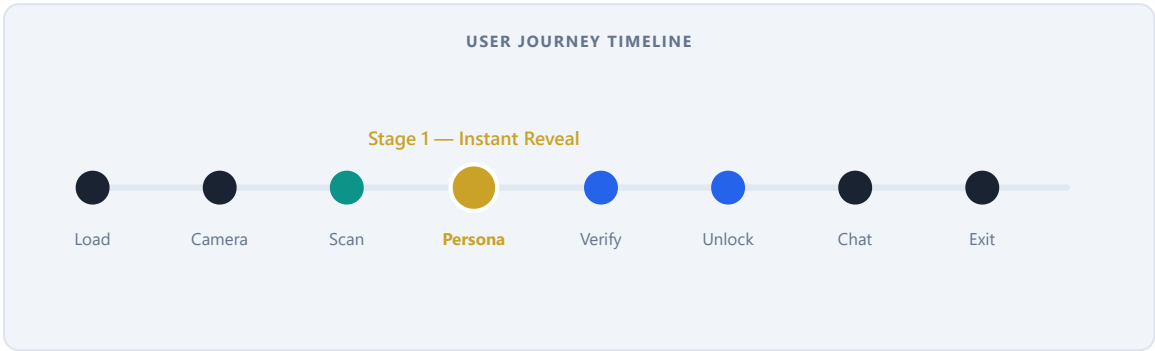
Axeraï — Per Product Interaction Budget

Own TTS server + Gemini Flash-Lite (approx)

Budget	Voice chars	Gemini tokens	Talk time
₹2	4k	59k	12 min
₹3	6k	88k	18 min
₹4	8k	1.2L	24 min
₹5	10k	1.5L	30 min sweet spot ~30 min
₹6	12k	1.8L	36 min
₹7	14k	2.1L	42 min
₹8	16k	2.4L	48 min
₹9	18k	2.6L	54 min
₹10	20k	2.9L	60 min
₹12	24k	3.5L	72 min
₹15	30k	4.4L	90 min

Voice ~4000 chars/₹1 • ~50% voice + 50% Gemini • Future own GPU ~₹50k/mo

3 End-to-End User Journey



PHASE BREAKDOWN

Intro & Permissions

Asset preload (FBX, MindAR target, card video). Camera on user gesture. Back camera preferred.

AR Session

MindAR tracks packaging target. Product video on anchor plane. Persona mounts same anchor.

Conversation

Live mic STT → Gemini reply → Inworld TTS. Mic released while AI speaks. Tap-to-fix on transcript.

Session End

Summary + KNOWN FACTS saved. Axerai AI tokens + voice tokens logged per code.

Axerai — NOW: Inworld + Gemini Flash-Lite

Per product interaction cost (approx, ₹96.5/\$1)

Talk time	Voice chars	Gemini tokens	Total cost
12 min	4k	59k	~₹11
18 min	6k	88k	~₹16
24 min	8k	1.2L	~₹21
30 min	10k	1.5L	~₹27 SWEET SPOT
36 min	12k	1.8L	~₹32
48 min	16k	2.4L	~₹43
60 min	20k	2.9L	~₹53
90 min	30k	4.4L	~₹80

🔊 Inworld TTS-2 \$25/1M • Gemini Flash-Lite • ~5x vs own server

Gemini + Inworld per-turn cost model

4 Three-Layer Verification Architecture

Core logic: Layer A is the fast visibility engine — persona appears immediately when product is detected. Layers B and C run in the **background** and return true/false without blocking Stage 1 reveal. One verify cycle per session.

Layer A
FAST

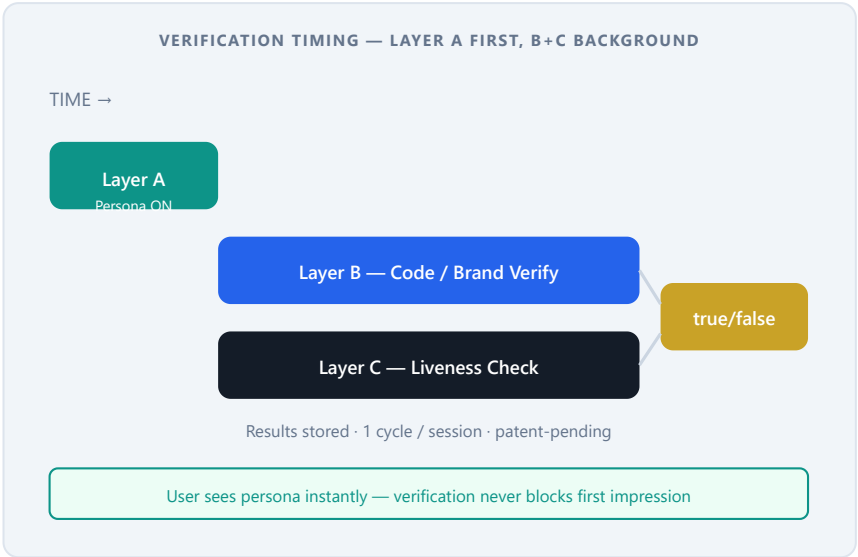
Product Visibility Engine
Trigger: Physical product / image target visible in camera.
Action: Instantly reveal 3D persona on packaging anchor (Stage 1).
Output: Persona live · AR session active.

Layer B
BG

Product Code / Brand Verification
Trigger: After Layer A — background Gemini vision on captured frame.
Action: Verifies brand text / product code (e.g. **AXE-327T5**).
Output: **true** = authentic · **false** = wrong/missing.

Layer C
BG

Liveness / Anti-Spoof
Trigger: Parallel with Layer B — background.
Action: Detects real product vs photo-on-screen, monitor replay, printed spoof.
Output: **true** = live product · **false** = spoof detected.



Axerai — 10 Brands Scale

3–5 million interactions / year (perfume, shoes, jewellery, lifestyle)

3M / year	5M / year
Per brand 3L scans ~25k / month	Per brand 5L scans ~42k / month
Own TTS ~13 GPUs (~₹6.5L / mo) or lighter 12-min avg ~5 GPUs (~₹2.5L / mo)	Own TTS ~21 GPUs (~₹10.5L / mo) or lighter 12-min avg ~9 GPUs (~₹4.5L / mo)
Inworld ~₹8 crore	Inworld ~₹13.5 crore

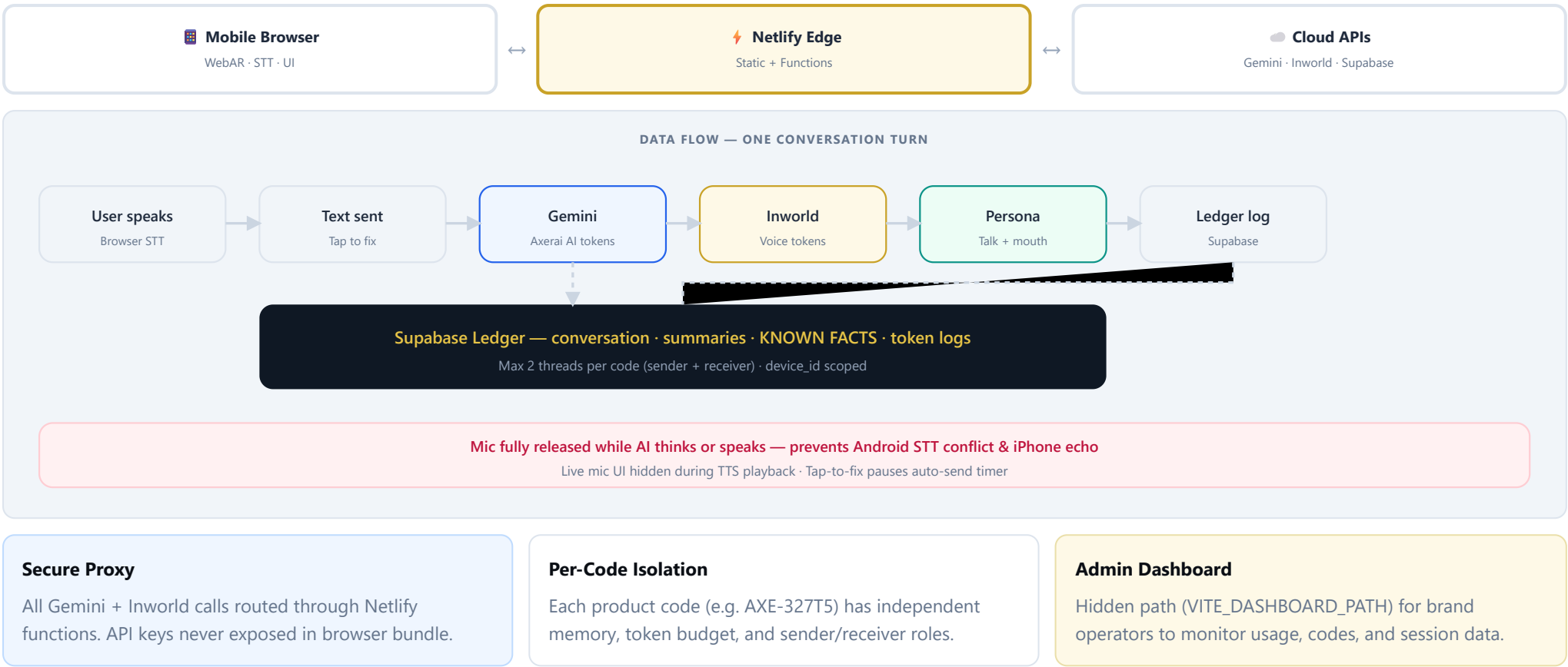
1 interaction
~ 30 min (₹5-slot)

lighter AR ~ 12 min
cuts GPUs ~half

Future own TTS wins
vs Inworld **crores**

Page 5

5 System Architecture



6 D3 Character — Animation & Lip Sync

MESH

FBX Skinned

Mixamo-style rig on AR anchor. Scale 0.008 · position tuned to product packaging.

IDLE

idle.fbx

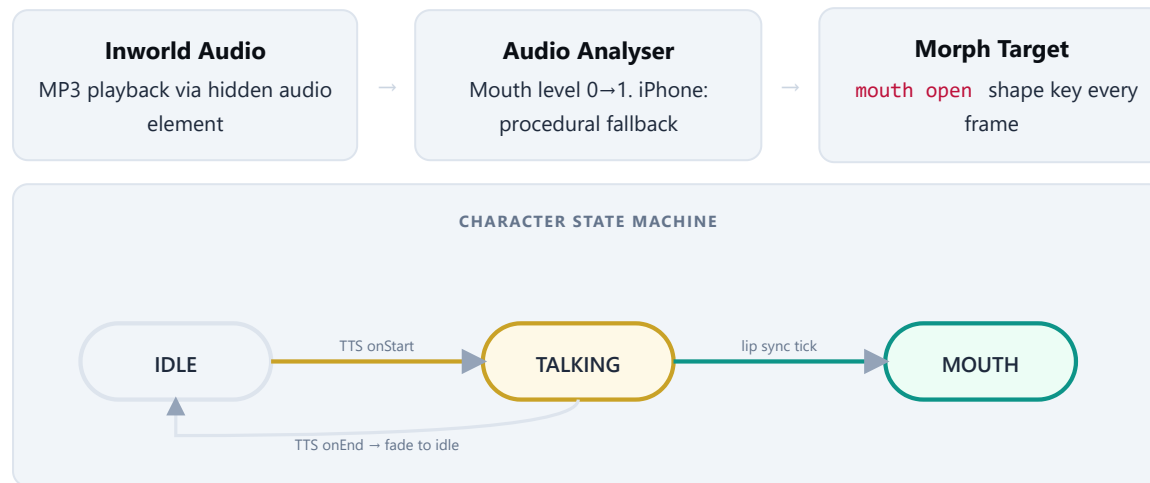
Standing loop when persona silent. Crossfade 0.35s — no T-pose flash on transition.

TALK

talking 4/5/6

Random talk clip while speaking. Switches only when audio actually starts playing.

MOUTH OPEN / CLOSE PIPELINE



7 AI Brain — Gemini

ITEM	DETAIL
Primary model	gemini-3.1-flash-lite
Fallback	gemini-flash-lite-latest
Persona prompt	Hinglish output, memory rules, gift scenarios
Memory	KNOWN FACTS (LOCKED) from ledger
Call types	verify · welcome · chat · summary
Production	Netlify function proxy — key never in browser

8 Voice — Inworld

MVP voice: Inworld TTS with custom cloned brand persona. Browser speechSynthesis is fallback only.

Clean

 →

Inworld

 →

MP3

 →

Lip Sync

 →

Log

9 STT & Live Mic

FEATURE	IMPLEMENTATION
API	Web Speech API · en-IN
Auto-send	2200 ms silence after speech
Tap to fix	Edit transcript; timer paused
During TTS	Mic released · UI hidden
Android	No getUserMedia during STT

PHASE: 10 BRANDS TAGGED SKU (NOT 100% CATALOG)

#	BRAND	TAGGED SKU UNITS/DAY	SCAN RATE	₹/SCAN	SCANS/DAY	REVENUE/DAY	₹/BOX
1	Nike	50,000	10%	₹5	5,000	₹25,000	₹0.50

Future scale path — dedicated inference server economics

10 Ledger, Memory & Token Metering

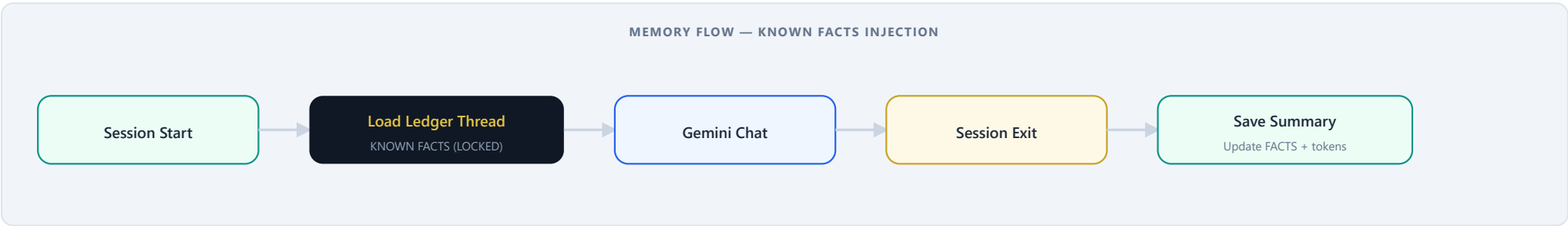
Table `ledger_threads` — maximum 2 rows per product code (sender + receiver).

COLUMN	PURPOSE
verification_code	Product code e.g. AXE-327T5
device_id	Stable browser ID
role	sender receiver
conversation	Full chat with session markers
session_summaries	Exit summaries + FACTS block
axerai_ai_usage	Axerai AI tokens (Gemini)
axerai_voice_usage	Axerai voice tokens (Inworld chars)

PER-CODE BUDGET PACKAGES (PILOT)

PACKAGE	TIME	AI TOKENS	VOICE TOKENS
Light	~12 min	~59,000	~4,000
Standard	~30 min	~1.5L	~10,000
Deep	~60 min	~2.9L	~20,000

When ~200–300 Axerai voice tokens remain → persona speaks warm exit line → session ends for that code.



11 Deployment & Security

Environment

Local: VITE_GEMINI_API_KEY ·
VITE_INWORLD_ENABLED
Production: server-only GEMINI +
INWORLD keys via Netlify functions

Security

3-layer verify · one cycle per session · API
keys on server · per-code usage caps ·
patent-pending architecture

One-line definition: Axerai MVP transforms physical packaging into an intelligent conversational touchpoint — scan, verify, persona, voice, memory — per product code
AXE-327T5 .

12 Project Structure

FILE / FOLDER	ROLE
src/App.jsx	Main AR + chat orchestration
src/myraModel.js	D3 character load, animations, mouth morph
src/myraLipSync.js	Audio → mouth level
src/myraPrompt.js	AI persona system prompt + user prompts
src/myraLedger.js	Supabase memory + token logging
src/myraVerify.js	Layer B/C vision verification
netlify/functions/	Gemini + Inworld TTS secure proxies
supabase/fresh_start.sql	Full database schema
public/models/	FBX character + animation assets

